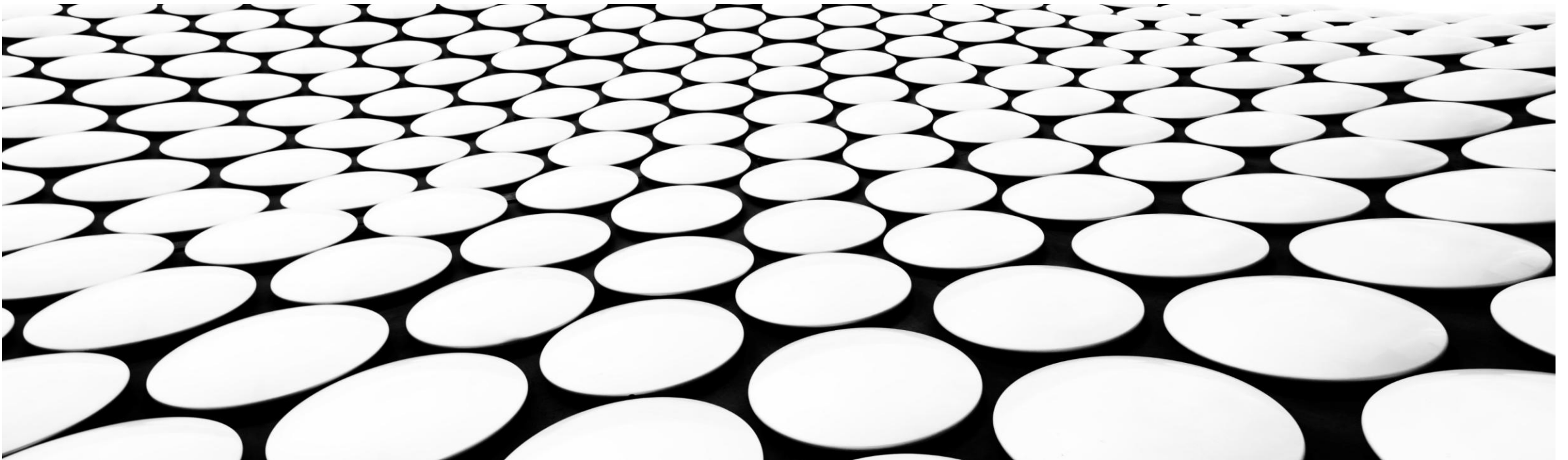
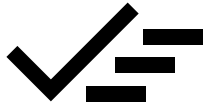


DATA COLLECTION

Comparing approaches



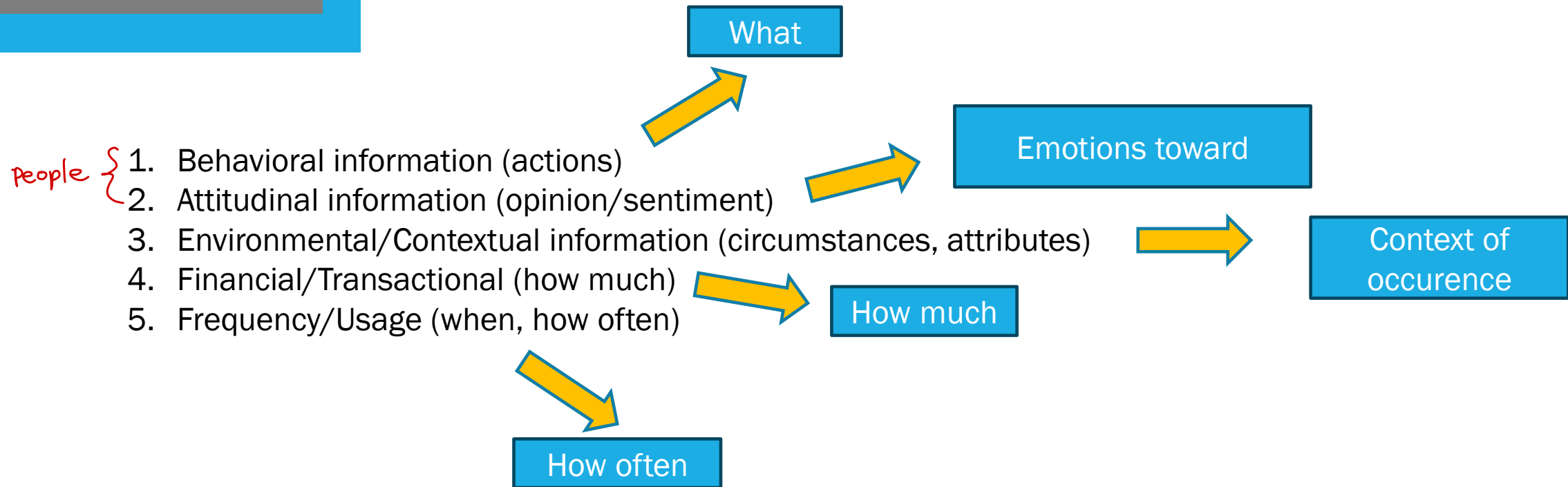


DATA COLLECTION

- Decision Factors for Data Collection Method Selection
- Three comparative case studies

DATA COLLECTION

What are we trying to capture?



Self-Reported Data	Surveys Interviews Focus groups
Data directly provided by individuals	
Behavioral and Observational Data	Observations Experiments
Data collected through monitoring or observing human behavior.	

Automated Data	Logs Sensor Transactions
Data generated obtained through an automated system.	
Digital and Platform-Based Data	Social Media Data Web Scraping APIs
Data originating from digital platforms, systems, or online activity.	

DECISION FACTORS



Quite overwhelming... How to choose/decide on data collection method?

Resource constraints
(Effort/Cost)

Time constraints

Goal of the study

Availability of already collected data

Domain expertise

Technical expertise

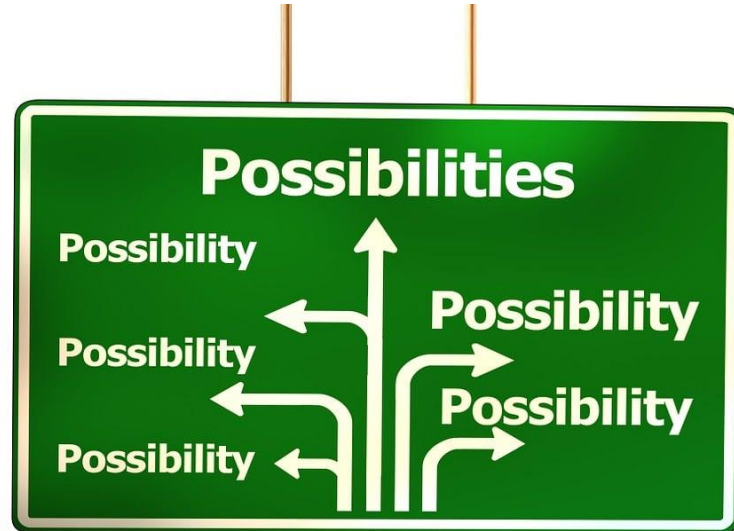
Data storage possibilities

Ease of use and integration

Reliability

Ethical and Privacy concerns

Data Collection



1. Purpose (Problem Statement/Question/Goal of Study)
2. Resources (Time/Effort/Cost)
3. Availability of already collected data
4. Domain Expertise
5. Technical Expertise
6. Data Storage possibilities (cost)
7. Reliability
8. Ease of use (feasibility) and integration
9. Ethical and Privacy concerns



1. Purpose

Which collection method would allow us to answer our research question?

Easy

What is the expected bounce rate if the company adopts the suggested website redesign?

A/B Testing

Not so easy

How much money are younger people spending on clothing compared to older people?

Survey
Observation
Logs
Transactions
Social Media

2. Resources

How much money, time, equipment do we have?

Not so easy

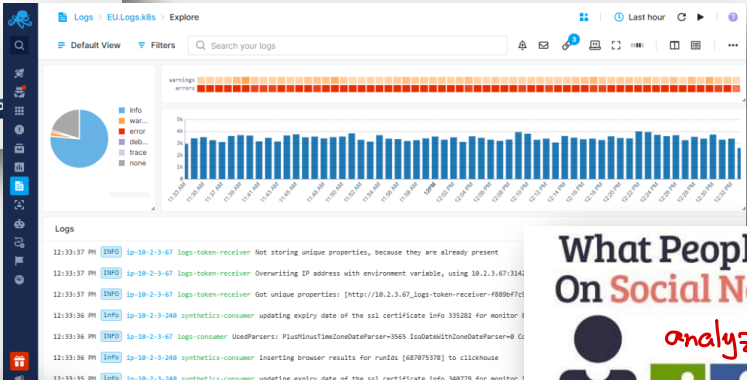
How much money are younger people spending on clothing compared to older people?

Survey
Observation
Logs
Transactions
Social Media

Survey
Observation
Logs
Transactions
Social Media



surveys



logs

Observational Research Methods

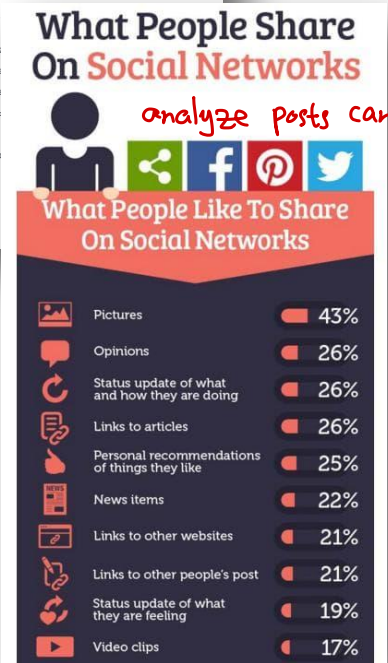
- Have a Clear Objective
- Get Permission
- Unbiased Observation
- Hide Your Observers
- Documentation
- Data Analysis

QuestionPro

observational

	A	B	C	D	E	F
	Date	Transaction ID	Customer Name	Category	Amount	Payment Method
1	2023-01-01 00:00:00	T00001	John Doe	Forex	54697.34	Check
2	2023-01-01 11:40:32	T00002	Alice Brown	Equity	-6230.63	Cash
3	2023-01-01 23:21:05	T00003	Jane Smith	Derivative	99837.92	Wire Transfer
4	2023-01-02 11:01:37	T00004	Jane Smith	Commodity	4473.36	Credit Card
5	2023-01-02 22:42:10	T00005	Jane Smith	Fixed Income	71438.19	Cash
6	2023-01-03 10:22:42	T00006	Bob Johnson	Forex	80311.67	Credit Card
7	2023-01-03 22:03:15	T00007	Jane Smith	Forex	31036	Check
8	2023-01-04 09:43:47	T00008	Charlie Green	Forex	11653.73	Wire Transfer
9	2023-01-04 21:24:19	T00009	John Doe	Derivative	868.59	Wire Transfer
10	2023-01-05 09:04:52	T00010	Alice Brown	Commodity	72346.66	Credit Card
11	2023-01-05 20:45:24	T00011	Alice Brown	Forex	39791.89	Credit Card
12	2023-01-06 08:25:57	T00012	John Doe	Commodity	68508.95	Wire Transfer
13	2023-01-06 20:06:29	T00013	Charlie Green	Fixed Income	90694.84	Credit Card
14	2023-01-07 07:47:02	T00014	Bob Johnson	Equity	6124.21	Wire Transfer
15	2023-01-07 19:27:34	T00015	Alice Brown	Equity	91108.81	Online Banking
16	2023-01-08 07:08:06	T00016	Bob Johnson	Fixed Income	35278.91	Check
17	2023-01-08 18:48:39	T00017	Bob Johnson	Equity	23579.37	Wire Transfer
18	2023-01-09 06:29:11	T00018	Alice Brown	Forex	93736.85	Check
19	2023-01-09 18:09:44	T00019	Bob Johnson	Derivative	98971.69	Check
20	2023-01-10 05:50:16	T00020	John Doe	Equity	11878.14	Online Banking
21	2023-01-10 17:30:49	T00021	Jane Smith	Fixed Income	62252.22	Wire Transfer
22	2023-01-11 05:11:21	T00022	Alice Brown	Derivative	1714.48	Credit Card
23	2023-01-11 16:51:54	T00023	Jane Smith	Fixed Income	61600.54	Credit Card
24	2023-01-12 04:32:26	T00024	Alice Brown	Forex	81004.46	Wire Transfer
25	2023-01-12 16:12:58	T00025	Charlie Green	Fixed Income	65294.84	Wire Transfer
26	2023-01-13 03:53:31	T00026	Jane Smith	Forex	35906.65	Check
27	2023-01-13 15:34:03	T00027	Alice Brown	Equity	32137.3	Check
28	2023-01-14 03:14:36	T00028	Bob Johnson	Derivative	33243.47	Cash
29	2023-01-14 14:55:08	T00029	Jane Smith	Commodity	54868.3	Check
30						

transactions



social media

requires technical expertise

3. Availability of already collected data

Is there already available data to help us answer our research question?

we need to evaluate the research question

Types of Data Sources: A Comprehensive Guide to Understanding Different Data Sources



Sumana Dotnettricks · Follow
10 min read · Jun 13, 2023

In this article, they also highlight secondary data

Secondary or Pre-existing Data	Government Databases Academic Research Industry Reports Publicly Available Data Media Sources
Pre-existing data collected by someone else for a different purpose	

Statistics Canada

Government Database

The United States Census Bureau's database offers comprehensive demographic data, allowing researchers to examine population trends and patterns across different regions and time periods.

Publicly Available Data

The World Bank's Open Data initiative provides free access to a vast repository of global economic indicators, enabling researchers and policymakers to track and analyze economic trends.

Replace the
collection method

Is there secondary data that contains **sufficient information** to answer our research question?

Replace part of the
collection method

Is there secondary data that contains some information that would contribute to answer our research question?

Provide additional
context for study

Is there secondary data that complements the collection data (e.g. providing additional context) ?

Example of Loyalty program of clothing store. When signing customers might have included their age and/or their postal code. Different aspects of demographics can then be inferred from postal code, as published by some government (e.g. provincial, municipal) data.

4. Domain Expertise

Do we (anyone in our team) know about the area of study?

Environment: Air Quality

How do air quality levels vary across different regions of Ottawa during peak traffic hours?



Behavioral (non human!)
Which sensors to use?

Transportation Services

What are the most common complaints about public transportation services in Gatineau?



Opinion (very human!)
Which questions are
targetting the issues?

5. Technical Expertise

Do we (anyone in our team) have technical expertise in line with the collection method?

What is the expected bounce rate if the company adopts the suggested website redesign?



Can you design and put in place A/B Testing?

How do social media users perceive the new health policy announced by the government?



Can you design and put in place Information Extraction from text found in Social Media?

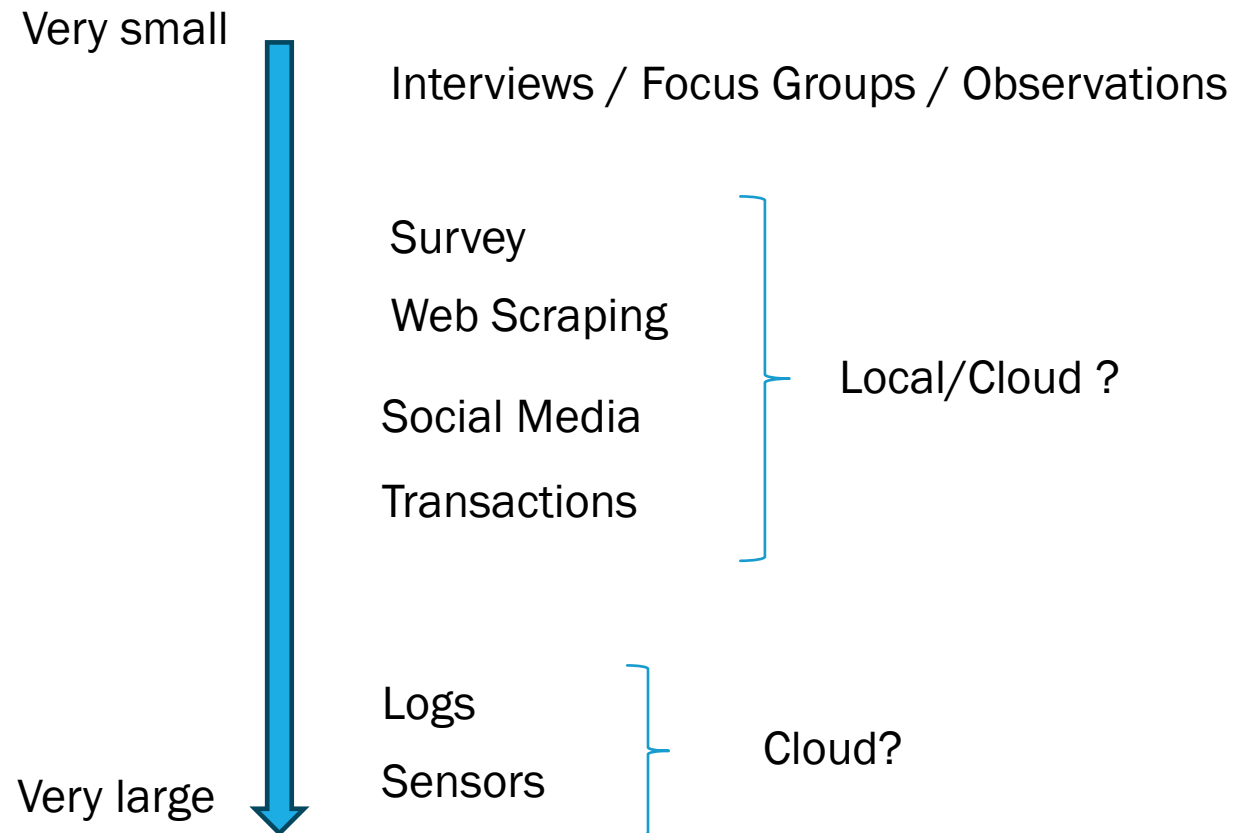
What are the primary motivations of employees for choosing remote work over on-site work in a company?



Can you design and put in place one-on-one interviews or focus groups to obtain self-reported data?

6. Data Storage possibilities

What is our budget (and current equipment) for local and/or cloud storage?



7. Reliability

Do we have confidence in the implementation of the collection method and the obtained answers?

Think of one or two potential issues of each approach?

	Data Collection	Possible reliability issue
Self-Reported Data	Survey	Poorly written questions. Users providing impressions based on memory, not necessarily factual.
	Interviews	Interviewer not knowledgeable enough. Participants not cooperative.
	Focus Groups	Moderator not experienced. Participants not expressing their real opinion.
Observational and Behavioral	Observation	Observation is too intrusive. No participant or reluctant participants.
	Experiment (A/B)	A/B setup is incorrect (e.g. random not really random). Controlled variable not well identified.

	Data Collection	Possible reliability issue
Automated	Logs	Logging platform not functioning well. Turnover/buffer (amount of data kept) is not sufficient.
	Sensors	Sensors malfunctioning.
	Transactions	DB not set up properly.
Digital and Platform based	Social Media	Extraction method is imperfect.
	Web Scraping	Extraction method is not adapted to new version of web site.
	APIs	Data provided not from a trustable source.

8. Implementation and integration

Is the collection method « implementable » (feasible) to answer the research question? If so, will the data collected be easily usable?

Question: What are the primary motivations of employees for choosing remote work over on-site work in a company?

Implementation --- Feasibility of the actual collection method



Sensors are just not possible

Integration – Once it's collected, can we use it?



What to do with interviews conducted? What can be said?

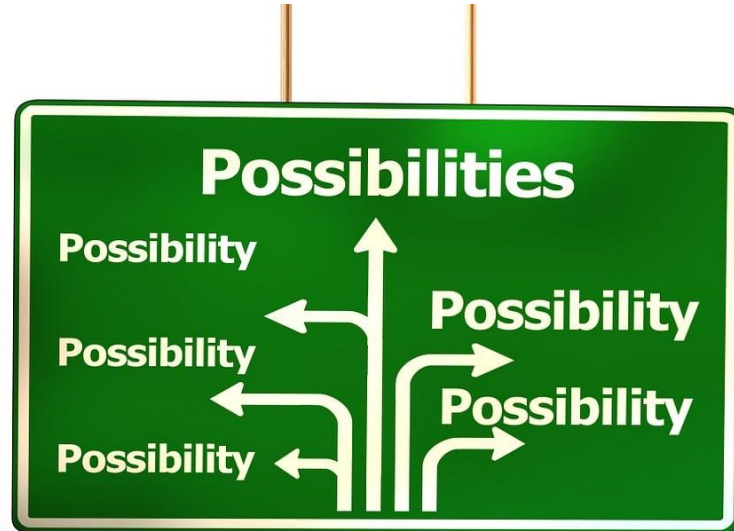
9. Ethical and Privacy concerns

Will the collection method infringe on people's privacy and create ethical concerns?

How much money are younger people spending on clothing compared to older people?

Data Collection Method	Example	Ethical concern
Survey	I provide my age and how much I spend per month on clothing.	I do it from my own will.
Observations	I'm being observed in a clothing store.	Did I give my permission for that?
Logs	I'm being « followed » through my online searches for various items to buy.	Passive observation (trickier... cookies everywhere). E.g. Google knows where I live and what I click on.
Transactions	I buy clothing with my credit card.	My credit card transactions would be linked to my age/clothing spending. Have I agreed to the use of aggregated data?
Social Media	I am a follower of a store to know their new collections and sometimes comment on them.	It's a bit like web scraping... legal (shared content), but what about « friends » or « followers »?)

Data Collection



1. Purpose (Problem Statement/Question/Goal of Study)
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Case Studies

Case Study Commuting Habits

Goal: Understand how people commute and why they choose certain transportation modes.

Case Study Eating Habits

Goal: Understand both *what people eat* and *why they eat that way*.

Case Study Energy Consumption

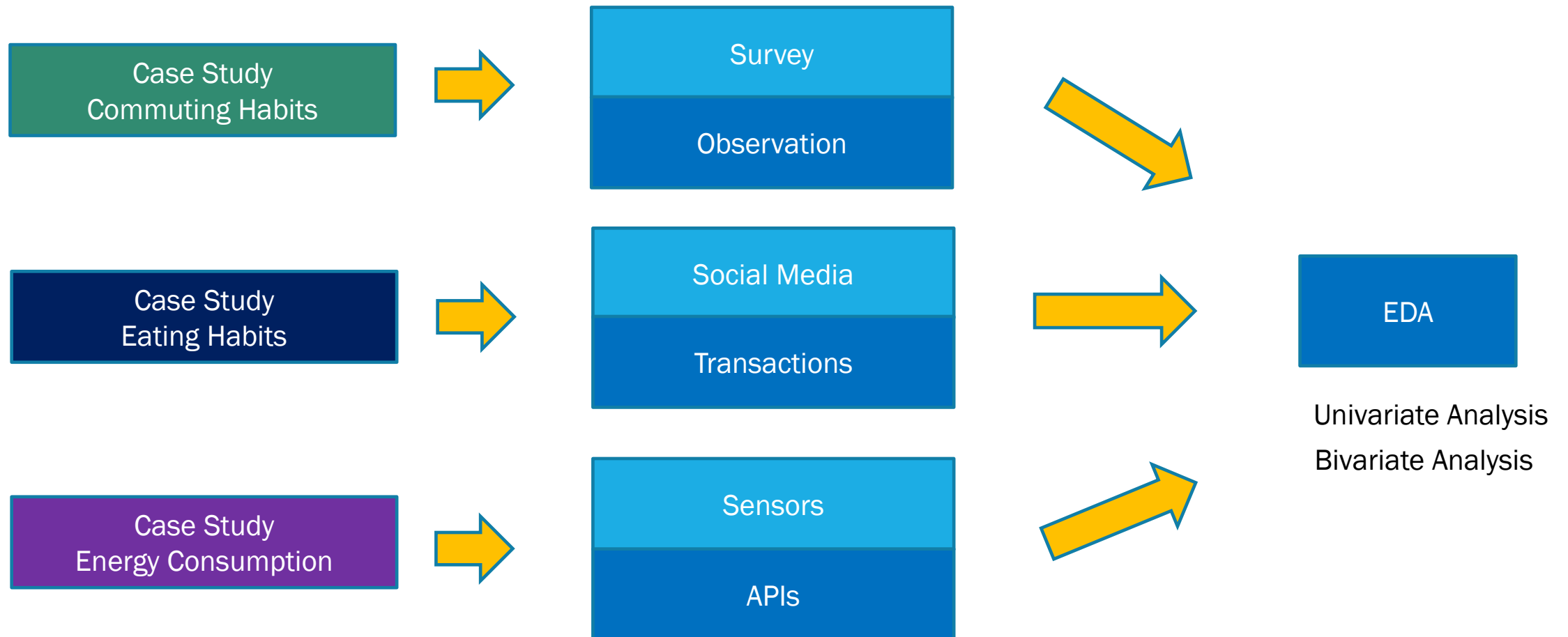
Goal: Understand *energy use behavior, perceptions, and influencing factors*.

Smaller

Very Large

Case studies:

Focus on two collection methods.



Case Study

Commuting Habits

Survey

Observation

Survey

Case Study Commuting habits

Impact on EDA –
Numerical/Categorical

Section 2: Commuting Patterns

4. What is your primary mode of commuting?

- ☐ Car
- ☐ Bus
- ☐ Train / Subway
- ☐ Bicycle
- ☐ Walking
- ☐ Other: _____

5. How long is your average one-way commute?

- ☐ Less than 15 minutes
- ☐ 15–30 minutes
- ☐ 31–45 minutes
- ☐ 46–60 minutes
- ☐ More than 60 minutes

Section 1: Demographics

1. Age: _____

2. Gender:

- ☐ Male
- ☐ Female
- ☐ Non-binary / Third gender
- ☐ Prefer not to say

3. Employment/Study Status:

- ☐ Full-time employee
- ☐ Part-time employee
- ☐ Student
- ☐ Other: _____

Often request for
demographics

Section 3: Preferences & Decision Factors

8. Which factor is most important when choosing your commuting mode? *(Select up to 2)*

- ☐ Cost
- ☐ Travel time
- ☐ Convenience
- ☐ Environmental impact
- ☐ Comfort
- ☐ Safety
- ☐ Other: _____

9. Do you consider environmental impact when choosing your commuting mode?

- ☐ Always
- ☐ Sometimes
- ☐ Rarely
- ☐ Never

10. Would you be willing to change your commuting mode if it saved money, time, or was more environmentally friendly?

- ☐ Yes, definitely
- ☐ Maybe
- ☐ No



Does there seem to be a bias here?

Section 4: Behavior & Technology

11. Do you use apps or services to plan your commute?

- ☐ Yes
- ☐ No

12. If yes, which apps or services do you use? (Open-ended)

• _____

13. Do you track your commute time or expenses?

- ☐ Yes, regularly
- ☐ Occasionally
- ☐ No

Optional Section 5: Open Feedback

14. Please share any comments on how commuting could be improved in your area:

• _____

Open-ended questions.
Text Analysis.

Observations

Case Study
Commuting habits

Observational Setting

- Location: Downtown intersection near transit hub
- Time window: 7:30–9:00 AM (weekday)
- Method: Manual counts every 15 minutes

Date	Time Block	Cars	Buses	Bikes	Pedestrians
2025-03-04	7:30–7:45	128	6	14	52
2025-03-04	7:45–8:00	162	8	21	68
2025-03-04	8:00–8:15	174	9	25	74
2025-03-04	8:15–8:30	158	7	19	63
2025-03-04	8:30–8:45	141	6	17	58

Comparison / Integration

	Survey	Observation
What is measured?	Self-reported behavior and perceptions	Directly observed commuting behavior
Whos is represented?	Survey respondents	People passing through a specific location
Level of details	Individual-level	Aggregate
Temporal coverage	Typical or average behavior (weekly)	Specific dates and time windows
Spatial coverage	Broad (city)	Narrow (intersection)

Case Study

Eating Habits

Social Media

Transactions

Social Media

Case Study Eating Habits

Post 1 (X / Twitter)

"Groceries are wild lately. \$70 and somehow still nothing to cook 🤔"

Post 2 (Instagram story caption)

"Made a salad for lunch today... then ordered fries at 4pm 🍟"

Post 3 (Reddit – r/EatCheapAndHealthy)

"I want to eat better but by the time I get home I just don't have the energy to cook. Anyone else?"

Post 4 (TikTok comment)

"All these 'healthy meals' videos use ingredients I can't even find where I live."

Post 5 (Facebook neighborhood group)

"Any decent takeout places around here that aren't crazy expensive?"

Post 6 (X / Twitter)

"I swear meal prepping only works if you don't mind eating the same thing 5 days in a row."

Posts could provide more info that can be structured

But... the text is tricky to structure

Target Field	Example
PostID	SM01
Platform	X, Instagram, TikTok, Reddit
Timestamp	2025-02-25
Text	"Groceries are ridiculous lately. \$80 and I still need dinner ideas."
Likes	124
Shares / Retweets	18
Hashtags	#groceries, #inflation

Regular Expressions

- Extracting anything « regular » in its surface form
 - Prices
 - Dates/Times
 - Hashtags

Post 1 (X / Twitter)

"Groceries are wild lately. \$70 and somehow still nothing to cook 🤔"

Post 2 (Instagram story caption)

"Made a salad for lunch today... then ordered fries at 4pm 🍟"

Sentiment analysis

- Classify posts into Negative/Positive/Neutral

"Fast food is cheap but leaves me feeling guilty. Taste is okay."

"The grocery delivery app is easy to use, but the delivery was late."

But we should really talk
about Aspect-Based
Sentiment Analysis

Text

"Groceries are ridiculous lately. \$80 and I still need dinner ideas. #groceries #inflation"

"Salad for lunch today 🥗 small wins. #healthyeating #lunch"

"I want to cook more but I'm exhausted. #homecooking #busy"

"Meal prep looks great until day 3... #mealprep #reallife"

"Fast food is cheaper than cooking right now... #fastfood #costofliving"

"Trying the new vegan restaurant. Love the taste! #vegan #yum"

"My coffee machine broke. Now mornings are stressful. #coffee #fail"

"I finally organized my pantry, so much easier to cook now! #organization #mealprep"

"Delivery was late again... but food was still hot. #takeout #frustration"

"Cooking with kids is fun but messy! #family #cooking"

"I'm saving money by buying in bulk. #groceries #budget"

"Trying intermittent fasting, mornings are rough though. #fasting #health"

1. Automatically identify the Aspects

Example aspects for eating habits posts:

- Food cost / budget
- Food quality / taste
- Convenience / time
- Cooking / meal prep
- Service / delivery
- Kitchen / tools
- Health / diet
- Experience / stress / enjoyment

PostID	Food cost	Food quality	Convenience	Cooking/M meal prep	Service/Delivery	Kitchen/Tools	Health/Diet	Experience/Stress
SM01	Negative	Neutral	Negative	Negative	NA	NA	NA	Negative
SM02	NA	Positive	Positive	NA	NA	NA	Positive	Positive
SM03	NA	NA	Negative	Positive	NA	NA	NA	Negative
SM04	NA	Positive	Neutral	Positive	NA	NA	NA	Neutral
SM05	Positive	Neutral	Positive	NA	NA	NA	Negative	Negative
SM06	NA	Positive	NA	NA	NA	NA	Positive	Positive
SM07	NA	NA	NA	NA	NA	Negative	NA	Negative
SM08	NA	NA	Positive	Positive	NA	Positive	NA	Positive
SM09	NA	Positive	NA	NA	Negative	NA	NA	Neutral
SM10	NA	Positive	NA	Positive	NA	NA	NA	Positive
SM11	Positive	NA	Positive	NA	NA	NA	NA	
SM12	NA	NA	Negative	NA	NA	NA	Positive	

Sentiment classification
per aspect + structuring

Transactions

Case Study
Eating Habits

TransactionID	Date	StoreType	ItemDescription	Category	Qty	TotalCost
845210	2025-02-28	Grocery	Rotisserie Chicken	Prepared Foods	1	9.99
845211	2025-02-28	Grocery	White Bread (Store)	Bakery	1	2.49
845212	2025-02-28	Grocery	Bananas	Produce	6	1.86
845490	2025-03-01	Fast Food	Large Fries	Side	1	4.29
846102	2025-03-02	Delivery App	Chicken Shawarma Plate	Restaurant Meal	1	16.75
846330	2025-03-03	Grocery	Frozen Lasagna	Frozen Meals	1	6.99
846998	2025-03-04	Coffee Shop	Latte (Medium)	Beverage	1	5.25

Comparison / Integration

	Social media	Transactions
Type of data	Text (unstructured) and more	Categorical/Numerical (structured)
Captures	Opinions (often extreme ones)	Actual purchases
Best for	Understanding motivation (do a selection and study)	Measure behavior
Required tools	Text analysis (imperfect)	Validation

Case Study

Energy Consumption

Sensors

APIs

Sensors

Case Study Energy Consumption

Sensor Types Included

1. Smart electricity meter
2. Smart thermostat
3. Indoor temperature sensor
4. Occupancy / motion sensor

Electricity Smart Meter (Whole-Home)

timestamp	meter_id	power_kW
2025-02-12 06:00:00	MTR_88421	0.8
2025-02-12 06:15:00	MTR_88421	1.1
2025-02-12 06:30:00	MTR_88421	1.6
2025-02-12 06:45:00	MTR_88421	2.3
2025-02-12 07:00:00	MTR_88421	3.0

Smart Thermostat

timestamp	thermostat_id	setpoint_C	indoor_temp_C	hvac_mode
2025-02-12 05:50:12	THM_33210	21.0	19.4	heating
2025-02-12 06:10:08	THM_33210	21.0	19.9	heating
2025-02-12 06:28:44	THM_33210	22.0	20.3	heating
2025-02-12 06:47:01	THM_33210	22.0	20.9	heating
2025-02-12 07:15:33	THM_33210	20.0	21.2	off

Sensor Types Included

1. Smart electricity meter
2. Smart thermostat
3. Indoor temperature sensor
4. Occupancy / motion sensor

Indoor Temperature Sensors (Room-Level)

timestamp	temp_sensor_id	room	temperature_C
2025-02-12 06:00:00	TMP_LR_091	LivingRoom	19.6
2025-02-12 06:05:00	TMP_LR_091	LivingRoom	19.8
2025-02-12 06:10:00	TMP_LR_091	LivingRoom	20.1
2025-02-12 06:15:00	TMP_LR_091	LivingRoom	20.4
2025-02-12 06:20:00	TMP_LR_091	LivingRoom	20.7

Occupancy / Motion Sensors

timestamp	motion_sensor_id	room	event_type	occupancy_detected
2025-02-12 06:02:11	OCC_BED_7712	Bedroom	motion	TRUE
2025-02-12 06:18:47	OCC_KIT_8894	Kitchen	motion	TRUE
2025-02-12 06:45:03	OCC_BTH_3341	Bathroom	motion	TRUE
2025-02-12 07:32:56	OCC_LR_4502	LivingRoom	motion	TRUE
2025-02-12 08:15:29	OCC_LR_4502	LivingRoom	no_motion	FALSE

APIs

Case Study
Energy Consumption

Weather APIs

API Example	Data Provided	Example
OpenWeatherMap / WeatherAPI	Temperature, humidity, wind speed, precipitation, conditions	Temp_C: -15.1, Humidity: 40%, Wind_kmh: 22, Precip_mm: 0.2, Condition: Snow
Weather.gov (US NOAA)	Hourly temperature, precipitation, storm alerts	Temp_C: 2, Precip_mm: 5, Snow_mm: 0
Meteostat	Historical weather for trend analysis	Average daily temp, max/min, precipitation

Various APIs

1. Weather APIs
2. Electricity Price APIs
3. Spatial (positioning) APIs
4. Calendar (events) APIs

Electricity Price APIs

API Example	Data Provided	Example
IESO (Ontario) / EIA (US)	Hourly market price of electricity, demand forecasts	Price: 0.12 CAD/kWh, Demand: 12,500 MW
Nord Pool	Day-ahead market prices (Europe)	Price: €0.095/kWh, Forecast demand: 4,300 MW

Various APIs

1. Weather APIs
2. Electricity Price APIs
3. Spatial (positioning) APIs
4. Calendar (events) APIs

Spatial (positioning) APIs

API Example	Data Provided	Example
Google Maps / OpenStreetMap	Building footprint, land use, nearby facilities	Residential, commercial, industrial density
City of Ottawa Open Data	Street, building type, energy efficiency rating	Building type: Apartment, Year built: 1980, Energy rating: B

Calendar (events) APIs

API Example	Data Provided	Example
Google Calendar API / Public Holiday APIs	National holidays, school breaks, daylight savings	Date: 2025-02-17, Holiday: Family Day
Event APIs (Ticketmaster, local city)	Major public events, sports, concerts	Event: Hockey Game, Date: 2025-02-10

Various APIs

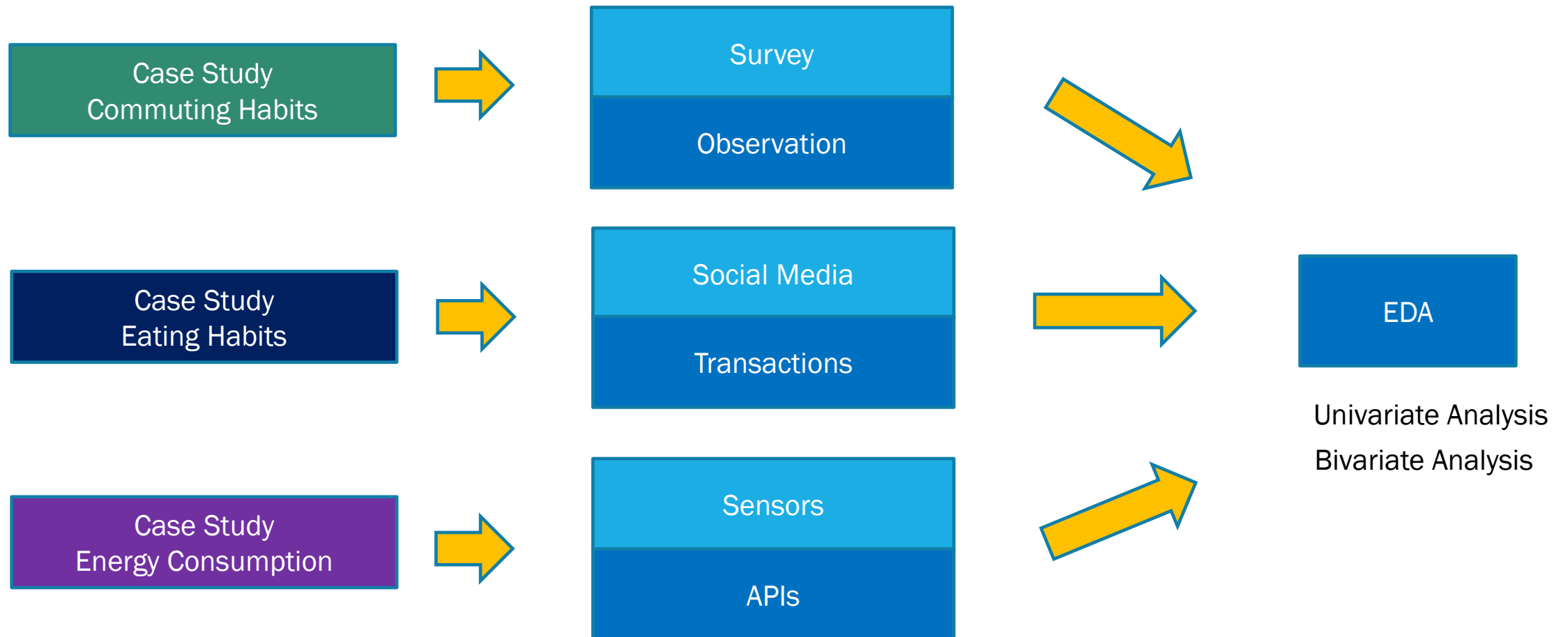
1. Weather APIs
2. Electricity Price APIs
3. Spatial (positioning) APIs
4. Calendar (events) APIs

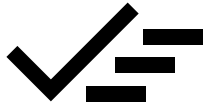
Comparison / Integration

	Sensors	APIs
What is measured	Energy usage, user actions, user behaviors	External conditions (weather, location, events)
Source	Physical devices in-home	External data providers (APIs)
Temporal coverage	Continuous (unless sensor defect)	APIs availability
Spatial coverage	Individual household	City, region

Case studies:

Focus on two collection methods.





DATA COLLECTION

- Decision Factors for Data Collection Method Selection
- Three comparative case studies